(i)

$$u = \frac{x-1}{x},\ dx = \frac{1}{(1-u)^2}\,du,\ x^{\frac{3}{2}}(x-1)^{\frac{1}{2}} = \frac{\sqrt{u}}{(1-u)^2}$$

$$\int \frac{1}{x^{\frac{3}{2}}(x-1)^{\frac{1}{2}}}\,dx = \int \frac{1}{\sqrt{u}}\,du = 2\sqrt{u} + C = 2\sqrt{\frac{x-1}{x}} + C$$

(ii)

$$\because \frac{d}{dx}\left(\sqrt{\frac{x+1}{x-2}}\right) = \frac{1}{2}\left(\frac{x+1}{x-2}\right)^{-\frac{1}{2}} \cdot \frac{-3}{(x-2)^2} = -\frac{3}{2(x-2)^{\frac{3}{2}}(x+1)^{\frac{1}{2}}}$$

$$\therefore \int \frac{1}{(x-2)^{\frac{3}{2}}(x+1)^{\frac{1}{2}}}\,dx = -\frac{2}{3}\sqrt{\frac{x+1}{x-2}} + C$$

(iii)

$$u = \frac{1}{x-1},\ x-2 = \frac{1-u}{u}$$

$$3x-2 = \frac{u+3}{u},\ dx = -\frac{1}{u^2}\,du,\ \begin{array}{c|c|c} x & 2 & \infty \\ \hline u & 1 & 0 \end{array}$$

$$\begin{aligned}
\int_2^{\infty} \frac{1}{(x-1)(x-2)^{\frac{1}{2}}(3x-2)^{\frac{1}{2}}}\,dx &= \int_1^0 \frac{-\frac{1}{u^2}}{\frac{1}{u}\cdot\sqrt{\frac{1-u}{u}}\cdot\sqrt{\frac{u+3}{u}}}\,du \\
&= \int_0^1 \frac{1}{\sqrt{(1-u)(u+3)}}\,du \\
&= \int_0^1 \frac{1}{\sqrt{4-(u+1)^2}}\,du \\
&= \left[\sin^{-1}\left(\frac{u+1}{2}\right)\right]_0^1 \\
&= \frac{\pi}{3}
\end{aligned}$$

This solution was submitted by **Jon**

Published and refined by **Jon**

Last updated: July 18, 2026